

Midterm 1 Review

Jacob Erlikhman

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Exercise 1 (*A free electron cannot absorb a photon*). In this problem, you will show that a free electron cannot absorb a photon and conserve both energy and momentum (hence, the photoelectric effect requires a *bound* electron).

Hint: Nothing is lost by working in the electron's rest frame, since the electron is massive and a process forbidden in one inertial frame is forbidden in all. In that frame the electron starts at rest while the photon has momentum p and energy $E_\gamma = pc$; recall $E_e = \sqrt{p^2c^2 + m^2c^4}$ for an electron of momentum p and mass m .

- Using momentum conservation, find the momentum and the energy of the electron just after it absorbs the photon.
- Now impose energy conservation (initial energy of the system equals final energy). Show that the two requirements can hold simultaneously only if $p = 0$, i.e. only if there was no photon at all, so a free electron cannot absorb a photon.

Exercise 2 (*Bohr Radius*). Let a be an estimate of the size of the hydrogen atom, so that the electron is confined to a distance of order a .

- Write down an expression for the momentum of the electron (order of magnitude estimate is fine).
- Supposing the orbit is circular, determine the centrifugal force on the electron.
- Equate this to the electrostatic force on the electron due to the proton, and solve for a to obtain

$$a = \frac{\hbar^2}{me^2},$$

which is the Bohr radius.

Exercise 3. Show that if $E < V(x)$ for all x , then $\psi(x) \equiv 0$.

Exercise 4. Suppose a free particle moves not on the x -axis but on a circle of radius R .

- Find the stationary states and the allowed energies.
- You should have found in (a) that there are degenerate states (i.e. linearly independent stationary states with the same energy). How is that consistent with the result of Exercise 2 of the Week 4 discussion section worksheet?

Exercise 5. Let $f(x)$ be an arbitrary function of position x and $g(p)$ be an arbitrary function of momentum p . Evaluate the commutators $[p, f(x)]$ and $[x, g(p)]$.

Hint: To compute the second commutator, you will want to first show that $x = i\hbar d/dp$.

Exercise 6. A particle of mass m sits in the ground state of an infinite square well with walls at $x = 0$ and $x = a$. At $t = 0$ the right wall is yanked *instantaneously* out to $x = 3a$. Because the move is instantaneous, the wave function has no time to change: an instant later it is still the old ground state,

$$\Psi(x, 0) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \quad (0 \leq x \leq a), \quad \Psi(x, 0) = 0 \quad (a < x \leq 3a),$$

but the stationary states are now those of the wider well, $\chi_n(x) = \sqrt{2/3a} \sin(n\pi x/3a)$, with energies $E_n^{\text{new}} = n^2\pi^2\hbar^2/18ma^2$. An energy measurement is then made.

- What is the probability of finding the particle in the ground state of the new well?
- What is the probability of finding it in the first excited state of the new well?
- Suppose instead the particle had started in the *first excited* state of the old well. What is the probability of finding it in the first excited state of the new well?
- Which new state is the single most likely outcome, and what is its energy? Comment on the result.

Exercise 7. A particle of mass m is in the harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$.

- Show that $\langle x \rangle = 0$ in all stationary states ψ_n .
- Show that $x^2\psi_0 = \frac{\hbar}{2m\omega} (\psi_0 + \sqrt{2}\psi_2)$, and find $\langle x^2 \rangle$ in the ground state.
- Find $\langle x^4 \rangle$ in the ground state, and show that it equals $3 \langle x^2 \rangle^2$.
- Find $\langle V \rangle$ and the standard deviation σ_V in the ground state.